

LAKSHITA BAGEJA

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PERSONAL DETAILS

Age : 25 years (as of Jan 1, 2026)

Nationality : Indian

Languages Known: English, French, Hindi

Linkedin : linkedin.com/in/lakshita-bageja-347641200

RESEARCH INTERESTS

Topological Quantum Field Theory, Anomalies and Index Theorems in Field Theory, Knot Theory in Chern-Simons, Brane Cosmology

EDUCATION

Integrated PhD (Physics) 2021 - present

Indian Institute Of Science Education And Research, Bhopal

M.S.(Physics) 2021 - 2024

Indian Institute Of Science Education And Research, Bhopal (7.46 CGP)

Bachelor of Science (Honors) - Physics
(Minor-Mathematics) 2018 - 2021

Sri Venkateswara College, University Of Delhi (8.3 CGPA)

Higher Secondary, CBSE (Class XII) 2018

Vivekanand School, D-Block, Anand Vihar, Delhi (93.8%)

SOFTWARE TOOLS

$\text{\LaTeX}$, Mathematica, Python, C++, Fortran

RELEVANT COURSE WORK

COURSES IN PHYSICS- Bosonic String Theory, Quantum Field Theory -I,II, Quantum Mechanics-I,II, Special and General Relativity, Decoherence And Open Quantum Systems, Statistical Mechanics, Nuclear And Particle Physics, Numerical Programming.

COURSES IN MATHEMATICS- Mathematical Methods of Classical Mechanics, General Topology, Algebraic Topology, Group Theory, Differential equations, Numerical Methods, Linear Algebra.

RESEARCH EXPERIENCE

Ongoing Project Jan'26 - present

Topic: Translation Invariant Defects

Description : Extension from Topological defects to translation invariant defects for various Ising CFT like Models

Ongoing Project Jan'25 - present

Supervisor: Dr. Nabamita Banerjee, Associate Professor, IISERB

Topic: Brane Cosmology in AdS/BCFT

Description : Cosmology-like model on EOW Branes in higher dimension AdS space using AdS/BCFT framework.

MS -Thesis

Aug'23 - April'24

Supervisor: Prof. Rajiv Gavai, Raja Ramanna Fellow , IISERB

Topic: Topological aspects of Schwinger Model at Finite Temperature

Description : Role of Homotopy and non-trivial topology in solving the 2-D EM Field Theory Toy Model on a closed spacetime manifold.

Summer Project

May'23 - Aug'23

Supervisor: Dr. Arnab Rudra, Assistant Professor, IISERB

Topic: 1-loop renormalization of Pure Yang-Mills

Description : Step by step computation of the Pure Yang-Mills one loop diagrams, checking of gauge invariance up to every order and then renormalizing the same in the MS Scheme.

iPhD Project

Jan'23 - April'23

Supervisor: Dr. Arnab Rudra, Assistant Professor, IISERB

Topic: Entanglement Entropy for Bi-partite Systems

Description : To investigate the Von-Neumann EE for bi-partite oscillator systems and understand it's extension to the multi-partite bosonic and fermionic systems.

Summer Project

May'22 - Aug'22

Supervisor: Dr. Arnab Rudra, Assistant Professor, IISERB

Topic: Symmetry and Degeneracy

Description : Explicit Symmetry Breaking pattern in various oscillator models and the effect of perturbations on the degeneracy of the system.

Dissertation Project

Jan'21 - April'21

Supervisor: Dr. Chandrabhan Dohare, Assistant Professor

Sri Venkateswara College, University Of Delhi

Topic: Optical Properties of MoS₂(Experimental Project)

Description: UV-visible, Ellipsometry and Raman study performed to analyze the origin of optical Excitonic Transitions, nature of vibrational modes and the number of layers of MoS₂ thin films.

Summer Internship Program (SRI-VIPRA 2020)

Jun'20 - Aug'20

Supervisor: Dr. Anand Pandey, Associate Professor

Sri Venkateswara College, University Of Delhi

Topic: Thermoluminescence Dosimetry

Description: Reading project on comparison of various materials for detection of radiation using thermoluminescence.

Summer Research Internship -2020

May'20

Supervisor: Dr. Abhishek Yadav, Assistant Professor, Aligarh Muslim University.

Topic: Heavy Ion Nuclear Interaction At Low Energies

Description: Reading project on study of dependence of Nuclear interaction on various parameters at low energies.

SCHOOL AND CONFERENCES ATTENDED

- 1. KMPB School 2025 : "D-branes in Physics & Mathematics"**
at Humboldt Universität zu Berlin, Germany
Presented Poster on : "Brane- Cosmology from AdS/BCFT" 22-26 Sep'25
- 2. Introductory School on Conformal Field Theory**
at IACS, Kolkata 18-22 Aug'25
- 3. Student Talk on Trending Topics in Theory(ST⁴) 2025, IISERB**
Short Talk on : "Cosmology from Branes in AdS" 14-26 Jul'25
- 4. Lecture Series: Charting topology of phase diagrams and field theories**
Role of Topology and Homotopy in Phase Transitions

- Speaker: Dr. Abhisodh Prakash, HRI** 18-20 Mar'25
- 5. In-house Symposium, IISERB**
Presented **Poster** on : "Witten's Anomaly and Topology" 3-4 Mar'25
- 6. National Strings Meet(NSM), IIT Ropar**
Presented **Poster** on : "Homotopy, Witten's Anomaly and Topology" 9-16 Dec'24
- 7. Lecture Series: SUSY in Condensed Matter**
Title: Aspects of Supersymmetry in Condensed Matter: from Frustrated Magnets and Superconductors to Topological Mechanics and Flat Bands.
Speaker: Dr. Krishanu Roychowdhury, SNIP 5-8 March'24
- 8. Regional Strings Meet, IISERB Chapter**
Prominent Speakers: Prof. Ashoke Sen, Prof. Shiraz Minwalla 18-20 Nov'22
- 9. Physics Lecture Series**
Title: The Real Symplectic Groups and their use in physics and Uncertainty Relations
Speaker: Prof. N. Mukunda 13-22 Sep'22
- 10. National Assembly of Researchers In Physics, IISERB** 25-26 Aug'22

TEACHING EXPERIENCE

PHY-304, Statistical Mechanics, T.A., IISERB	Jan'26- present
PHY-303N, PHY643, Quantum Mechanics, T.A., IISERB	Aug'25- Dec'25
PHY-304, Statistical Mechanics, T.A., IISERB	Jan'25- Jun'25
PHY-206, Physics Through Computational Thinking, T.A., IISERB	Jan'24- Jun'23
PHY-101, Mechanics, T.A., IISERB	Aug'23- Dec'23
11 Courses : AP Physics, AP Calculus - AB, BC(Part-time)	Aug'22- Jan'25
Position - Course Instructor	
Platform : Learntech, Online(40 hours each course)	

AWARDS AND ACHIEVEMENTS

- GATE- Physics(2023)** - Qualified with All India Rank (**AIR**) -**774**
- CSIR-JRF (Jun'2021)**- Qualified with **AIR-206**
Eligibility for Junior Research Fellow (JRF) by UGC-NET. Qualified for fellowship offered by CSIR
- IIT JAM- PHYSICAL SCIENCE (2021)**- Qualified with **AIR-315**

POSITION OF RESPONSIBILITY

- Organizing Committee of ST⁴2026 - Student conference 2026
- Departmental Representative(DR), PhD Physics, IISERB 2023-24
- Departmental Representative(DR), Integrated PhD Physics, IISERB 2021-22

CO-CURRICULAR ACTIVITIES

- Trekking** - Buran Ghati Trek, Himachal Pradesh -15,000 ft(2024), Phulara Ridge Trek, Uttarakhand-12,345 ft(2023)
- Athletics - 400m** - Runners up(2024), Intra-IISERB Sangharsh

3. **Marathon**- 13th,17th position in 5 kms mini marathon(2024,2023) at IISERB
4. **Volleyball** - Winner(Mixed (2023,2022) and Girls(2022)) Intra-IISERB Sangharsh
5. **Powerlifting** - Winner-2024, 2022 (50-60 kg) Intra-IISERB Sangharsh
6. **Debate** - 1st Prize in Tark-Vitark, 2021 at Hansraj College, University Of Delhi,
1st Prize in React-Chemical Dargal, 2020 Ramnarain Ruia College, Maharashtra